

CS3307 Group Project Stage #2

Raspberry Pi Smart Dashboard

User story & UML

Group Number: 19

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1. Introduction

In this stage of our project, we formalize the requirements by writing user stories, providing acceptance tests, estimating their complexity (in story points), and drafting an initial UML class diagram. We will also briefly discuss how these stories are tracked in Jira.

2. User Stories

User stories are grouped by **Required**, **Optional**, and **Wish-List** features. Each story follows the template:

- **User Story:** A concise statement in the form “As a [role], I want to [action] so that [benefit].”
- **Acceptance Tests:** The conditions or steps needed to verify that the feature works as intended.
- **Estimate (Story Points):** An approximate measure of effort/complexity. One story point is roughly one evening of focused development work for one person.

2.1 Required Features

2.1.1 Adaptive Screen Brightness & Wake Function

1. User Story

“As a user, I want the screen brightness to adjust automatically based on ambient light so that I can see the display clearly at all times.”

- **Acceptance Tests**

1. Place the device in a bright environment; verify that the screen brightness increases.
2. Place the device in a dark environment; verify that the screen brightness decreases.
3. Turn the ambient light sensor off in software; verify that brightness stays at a default level.

- **Estimate:** 5 story points

2. User Story

“As a user, I want the dashboard to wake up when I approach and go to sleep when I leave so that I can save power and reduce screen wear.”

- **Acceptance Tests**

1. Approach the dashboard from a few feet away; verify that the screen transitions from sleep/dim mode to active mode.
2. Remain out of range for a specified timeout; verify that the screen dims or turns off.

- **Estimate:** 5 story points

3. User Story

“As a user, I can manually put the dashboard to sleep using an on-screen button so that I can turn it off when not in use.”

- **Acceptance Tests**

1. Tap the on-screen “Sleep” or “Off” button; verify that the display turns off or dims.
2. Tap the screen or press a wake button; verify that the dashboard returns to active mode.

- **Estimate:** 3 story points

2.1.2 Media Playback & Gesture Navigation

1. User Story

“As a user, I want to play media (YouTube, Netflix, Spotify) so that I can watch videos or listen to music from my dashboard.”

- **Acceptance Tests**

1. Select a supported streaming service; verify that the login or authentication process is accessible.
2. Play a video or audio track; verify that playback controls (play, pause, volume) function correctly.
3. Check network status; verify that media playback gracefully handles slow or lost connections.

- **Estimate:** 5 story points

2. User Story

“As a user, I want to use a two-finger swipe gesture to quickly access the media playback interface so that I can switch between dashboard views efficiently.”

- **Acceptance Tests**

1. Perform a two-finger swipe from left to right; verify that the media playback interface slides into view.
2. Return to the main dashboard by swiping in the opposite direction; verify that the media interface slides out.

- **Estimate:** 3 story points

2.1.3 Raspberry Pi System Statistics

1. User Story

“As a user, I want to view my Raspberry Pi’s system statistics (CPU, RAM, storage, network) so that I can monitor its performance in real time.”

- **Acceptance Tests**

1. Open the “System Stats” screen; verify that CPU usage, RAM usage, and disk space are displayed.
2. Perform a CPU-intensive task (e.g., open media playback); verify that CPU usage updates accordingly.
3. Disconnect network or limit bandwidth; verify that network statistics reflect the change.

- **Estimate:** 3 story points

2.1.4 Touchscreen Gestures & Input

1. User Story

“As a user, I want to navigate the dashboard using intuitive touch gestures so that I can switch between screens and features easily.”

- **Acceptance Tests**

1. Swipe horizontally from the main screen; verify that the next screen (e.g., media or calendar) appears.
2. Swipe vertically from the main screen; verify that relevant menus or settings appear.
3. Attempt pinch/zoom on graphs or media; verify that it responds appropriately if supported.

- **Estimate:** 5 story points

2. User Story

“As a user, I want an on-screen keyboard to enter credentials or add events so that I can interact with the dashboard without a physical keyboard.”

- **Acceptance Tests**

1. Tap a text field; verify that the on-screen keyboard appears.
2. Type in text; verify that characters appear correctly in the text field.
3. Hide the keyboard; verify that the screen returns to the main layout without residual overlays.

- **Estimate:** 3 story points

2.1.5 Calendar Integration

1. User Story

“As a user, I want to view my calendar events on the dashboard so that I can stay informed about upcoming tasks.”

- **Acceptance Tests**

1. Open the “Calendar” screen; verify that a monthly or weekly view appears.
2. Check that existing events are visible on the correct dates/times.
3. Switch to a different month/week; verify that the view updates accordingly.

- **Estimate:** 3 story points

2. User Story

“As a user, I want to add and update calendar events directly from the dashboard so that I can manage my schedule easily.”

- **Acceptance Tests**

1. Tap a date/time slot; verify that an “Add Event” dialog appears.
2. Create a test event and save; verify that the new event is displayed on the calendar.
3. Edit an existing event; verify that changes are reflected immediately.

- **Estimate:** 5 story points

3. User Story

“As a user, I want to sync my dashboard calendar with Google Calendar so that all my events stay up to date.”

- **Acceptance Tests**

1. Authenticate with a Google account; verify that the dashboard can read events from Google Calendar.
2. Create an event on Google Calendar (on another device); verify that the new event appears on the dashboard after a sync.
3. Create an event on the dashboard; verify that it appears on Google Calendar.

- **Estimate:** 5 story points

2.1.6 Weather Station

1. User Story

“As a user, I want to see real-time weather updates so that I can plan my day accordingly.”

- **Acceptance Tests**

1. Open the “Weather” screen; verify that current temperature, humidity, and conditions (e.g., sunny, cloudy) appear.
2. Switch from indoor to outdoor data; verify that the display updates to show the correct information.
3. Turn off Wi-Fi; verify that an error or placeholder is displayed for outdoor weather data.

- **Estimate:** 3 story points

2. User Story

“As a user, I want to track indoor and outdoor temperature/humidity so that I can monitor my home environment.”

- **Acceptance Tests**

1. View the indoor sensor data; verify that it updates in real-time (e.g., changes in temperature if the sensor is heated).
2. Fetch the outdoor data from an external API; verify that the displayed data matches a known weather service.

- **Estimate:** 5 story points

3. User Story

“As a user, I want to view a 24-hour trend graph of temperature and humidity so that I can analyze environmental changes over time.”

- **Acceptance Tests**

1. Open the trend graph; verify that it shows the last 24 hours of data points for temperature and humidity.
2. Hover/tap on data points; verify that specific time and value appear.
3. Change date range to a different day; verify that the graph updates with new data (if implemented).

- **Estimate:** 8 story points

4. User Story

“As a user, I want to receive real-time weather advisories so that I can prepare for sudden weather changes.”

- **Acceptance Tests**

1. Trigger a mock weather advisory (e.g., a test alert from the API); verify that the dashboard shows a notification or alert.
2. Dismiss the advisory; verify that it no longer appears on the main screen.
3. Check that the advisory is archived or logged for future reference.

- **Estimate:** 5 story points

2.2 Optional Features

2.2.1 Bluetooth Connectivity

1. User Story

“As a user, I want to connect Bluetooth devices (e.g., keyboard, mouse, speakers) so that I can enhance my dashboard experience.”

- **Acceptance Tests**

1. Open the “Bluetooth Settings” panel; verify that the system scans for available devices.
2. Pair with a Bluetooth keyboard; verify that keystrokes appear in the dashboard’s text fields.
3. Pair with Bluetooth speakers; verify that audio output is routed correctly.

- **Estimate:** 5 story points

2. User Story

“As a user, I want to pair Bluetooth headphones so that I can listen to media privately.”

- **Acceptance Tests**

1. Open the “Bluetooth Settings” panel; verify that the headphones appear in the list of discoverable devices.
2. Connect to the headphones; verify that audio plays exclusively through them.
3. Disconnect headphones; verify that audio automatically reverts to the default speaker.

- **Estimate:** 3 story points

2.2.2 Customization Options

1. User Story

“As a user, I want to change my dashboard’s theme (light/dark/custom) so that I can personalize its appearance.”

- **Acceptance Tests**

1. Open “Settings” → “Appearance”; verify that theme options are displayed.
2. Select “Dark Mode”; verify that the entire dashboard switches to a dark color scheme.
3. Return to “Light Mode”; verify that the color scheme switches back.

- **Estimate:** 3 story points

2. User Story

“As a user, I want to enable a ‘Wallpaper of the Day’ feature so that I can see a new background image daily.”

- **Acceptance Tests**

1. Enable the “Wallpaper of the Day” option; verify that the current wallpaper updates to a new image (e.g., from NASA).
2. Check the next day; verify that a different wallpaper is displayed.
3. Disable the feature; verify that the wallpaper remains static.

- **Estimate:** 3 story points

2.3 Wish-List Features

2.3.1 AI-Powered To-Do List

1. User Story

“As a user, I want to dictate tasks using voice commands so that I can create a to-do list hands-free.”

- **Acceptance Tests**

1. Tap the “To-Do” icon; verify that a microphone button is present.
2. Press the microphone button and speak a task; verify that the system accurately transcribes the text.
3. Check that the new task is added to the to-do list upon completion of transcription.

- **Estimate:** 5 story points

2. User Story

“As a user, I want the AI to structure my tasks into clear, actionable points so that I can better organize my work.”

- **Acceptance Tests**

1. Speak a complex sentence describing multiple tasks; verify that the system splits them into separate bullet points.
2. Review the structured tasks; verify that each bullet point is grammatically clear and distinct.

- **Estimate:** 3 story points

3. User Story

“As a user, I want the AI to read my to-do list aloud so that I can review my tasks without looking at the screen.”

- **Acceptance Tests**

1. Tap the “Read Aloud” button; verify that the system uses text-to-speech to list all tasks.
2. Add or remove tasks while it’s reading; verify that it updates the list accordingly (if supported in real time).
3. Cancel the reading; verify that text-to-speech stops.

- **Estimate:** 3 story points

3. UML

Below is the high-level UML class diagram that captures our planned architecture. This diagram is not final and may evolve, but it reflects our current understanding of the classes, their responsibilities, and relationships.

